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Adaptive and Explainable Human-Robot Collaboration for Intelligent Sorting Systems: Woody the Educational Robot

Abstract

The present study proposes a novel integration of HRC, aiming at providing an inclusive platform for cognitive and sensory learning of differently abled students. The system consists of four key components, namely 1) a vision-based texture recognition component based on a self-supervised transformer (DINOv2) algorithm for object material identification, 2) a robotic learning component aimed at pattern recognition and linguistic ability improvement via object-based exercises, 3) a head and gaze-tracking object selection process that makes object interaction non-invasive due to the use of only one RGB camera and 4) an XAI-driven predictive sorting component that improves transparency in decision making when completing the task. By incorporating object detection and manipulation algorithms, attention tracking, and speech feedback, the system creates a multi-modal interactive and adaptive learning environment. Due to the presence of a human-in-the-loop component, a student can watch how the robot performs its task and gains additional knowledge about it. This way, the use of real-time feedback and explainability in robot decision making results in increased engagement, better understanding, and more trust.

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Keywords: Human–Robot Collaboration (HRC), Vision Transformers (DINOv2), Texture Recognition, Gaze Estimation, Head Pose Estimation, Object Detection (YOLO), Explainable AI (XAI), Adaptive Learning, Interactive Learning Systems, Pattern Recognition, Recommendation Systems

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1. Introduction

In this regard, human-robot collaboration (HRC) is an exciting area to explore in intelligent interactive systems. There is a need for adaptive and multi-sensory approaches in education for differently abled students apart from traditional approaches. The integration of robotics with computer vision and machine learning will create interactive environments where the student learns through observation and interaction.

Accurate material recognition is difficult in robotic sorting because of various textures and visual appearances [1],[2]. The system employs a vision-based texture recognition algorithm using DINOv2 [3] for classifying the materials into plastic, metal, glass, and paper with confidence scores. Region-of-interest segmentation is employed, and the object is moved to improve visibility.

Successful HRC should be flexible in adapting to the actions of the human and interacting in real-time [4],[5]. Here we present a robot-assisted learning system which is useful to differently abled children and normally developed children. It combines a vision-based pattern completion task with speech repetition [6].

Intuition plays a vital role as humans indicate their intentions using eye and head gestures [7]. The technology used includes a novel gaze and head pose-based object selection technique with the use of one RGB camera. Gaze and head pose information together with object detection enable the system to determine users' intentions, providing a solution without any specialized sensors, making the approach efficient and affordable [8].

Transparency and adaptability of the robotic behavior are critical to achieving positive effects when working with educational purposes [9]. The sorting policy used in the system is recommendation-driven; that is, actions are performed according to object properties detected in real-time. Moreover, explainability provided by the system in the form of XAI allows for understanding the robot's reasoning, which makes both teachers and pupils engaged [10]. As a result, it becomes possible for the latter to not only perform certain tasks but also analyze how they should be done. Driven by the lack of adaptive and interactive platforms for differently abled kids, the proposed solution integrates multiple techniques to support multi-modal learning.

2. Literature review

Sorting systems using robotics technology have made great progress due to the inclusion of artificial intelligence, which has increased their performance and adaptability. Nowadays, the state-of-the-art systems utilize deep learning-based object detection techniques, such as YOLO, for sorting and object manipulation [11]. Combining the use of RGB-D cameras and robotic manipulators facilitates the task of localizing and determining how to pick the objects [12]. Collaborative robots or cobots can be used to combine the strengths of robots and humans by facilitating the safe interaction of robots with people [13],[14]. However, these systems still face many obstacles, including the need for adaptability in different environments, efficient performance in dynamic situations, and clear reasoning processes [15],[16],[17]. In particular, in the field of special needs education, the learners may benefit from slow interactions, adaptability, and visual guidance for safer and more effective learning. Thus, any robotic sorting system should be adapted to special needs environments in order to accommodate learners of all abilities.

The implementation of IoT-based technologies is very important for developing capabilities of real-time perception and spatial awareness in robotic sorters. This is made possible by using RGB cameras which enable gathering information about object features, like its color, shape, and positioning. The use of this is fundamental to developing spatial representations and mapping out objects [18],[19]. Edge computing enables performing perception and decision-making tasks directly on the device, without having to rely on cloud technologies [20], thereby increasing the responsiveness of robotic sorters. Using multimodal inputs, which combine both physical and visual feedback is also beneficial for dealing with unstructured and changing environments [21]. It should be highlighted that such technologies and systems can be applied efficiently in educational purposes, especially when working with differently-abled students.

Material detection is a vital process in robot-based sorting machines, where material identification increases the performance of the sorting system and aids in user understanding [2],[22]. Material identification by analyzing visual features proves difficult due to factors such as changes in object surface textures, illumination conditions, and orientations [22]. Recently, advancements in deep learning algorithms, in particular, self-supervised learning using vision transformers like DINOv2, have made the process more efficient for visual feature analysis and textural

interpretation [3],[1]. Vision transformers can create effective representations from RGB image data that allow for the successful classification of materials like plastic, metal, glass, and paper [3]. Although most methods concentrate on industry use cases, there has been a growing trend in recent years of combining material recognition with intelligent user-oriented systems. In education, specifically for special needs students, robots are required to recognize the materials correctly while making sure they display them in a clear and manipulative manner [22],[23],[24].

Human robot interaction has been evolving into systems that are adaptive and user-centric, facilitating learning, therapy, and skill acquisition in various learning communities [4]. Current trends in educational robotics involve multimodal perceptual abilities such as object identification through visual means, communication through speech, and gestures [5]. Learning with the help of robots in patterned sequences facilitates increased attentiveness, completion of tasks, and better communication when learning involves human interaction with the system [6]. The use of collaborative robots with real-time vision and manipulation skills promotes cognitive involvement through interactive physical activity adjusted to individual performance [7]. Interaction through speech, verbal instructions, and patterned exercises help develop language skills, task comprehension, and learning corrections in cases where students require extra assistance [8]. Such characteristics make modern human-robot collaboration technologies well-suited for special education applications, where personalization, inclusion, and physical interaction are paramount [9].

The gaze and visual attention are key determinants of human intentions which can help to facilitate intuitive human-systems interactions. The use of the gaze as an interaction modality allows a means of deducing the focus of the learners without having to give any command [7]. Research work done in the field has considered integrating gaze in intention recognition and collaboration, where combining the gaze modality with AI planning has helped in predicting human actions [25]. The research works on gaze-based attention recognition show that task-related areas can be recognized by using the gaze of humans during collaboration in assembly [26]. Additionally, inference of human intentions from gaze is proven to be reliable when compared to other methods such as the use of EEG signals or motion cues [27]. New models include uncertainty modeling for gaze estimation, dual-attention processes, and 3D scene perception [28],[29],[30],[31]. Most of the systems developed so far depend on expensive sensor systems and multiple cameras, making them less practical, and they only deal with gaze and objects independently.

As robot autonomy increases, interpretability and explainability become crucial, especially in domains such as education and applications involving humans. Explainable AI helps explain why certain decisions are being taken and why, which boosts trust, safety, and usability [32],[33]. Explainability methods for deep learning include the use of SHAP or Grad-CAM to identify and highlight features contributing to decision-making [34],[35]; however, current methods usually only apply to individual actions or decision-making rather than whole sets of tasks, which is fundamental in robotics [36]. On the other hand, recommendation systems enhance task execution by offering optimized sequences of actions based on environment and object information [19],[37]. Hybrid models combine the benefits of logical rules with machine learning algorithms to achieve both flexibility and explanation [38],[39]. The combination of explainability in task recommendation systems in education tailored to special needs can provide insight into robotic behaviour from all involved parties.

3. Methodology

This system seeks to design a learning space where differently-abled and regular users can collaborate with robots through interactive, human-robot cooperative learning. This system uses material recognition, intent detection based on eye gaze tracking, prediction for sorting, and multimodal interaction. The proposed system enables safe, transparent, and inclusive adaptive learning through multimodal interaction, involving visual, tactile, and verbal modes of communication. For this purpose, two RGB cameras are deployed; one sensor is responsible for capturing the materials within the workspace for perception and prediction, while the second sensor detects the activities of the user in order to enable eye gaze and intent detection. Data preprocessing, including resizing, normalization, data augmentation, and segmentation of the background, helps ensure that the system remains robust to changes in illumination conditions and object positions.

The task of material recognition involves the use of a model based on DINOv2, where RGB images of materials such as plastics, metals, glasses, and papers are captured within a specified region of interest so as to reduce the effect of the background. The pre-processed images are then fed into the pre-trained DINOv2 model, where the extraction of visual features occurs, followed by material classification with certainty scores. The material class and object

locations are fed to the robot arm control system, where the object is placed in the right position to be clearly observed. This provides an opportunity for students to verify and/or learn the process of material recognition.

For interactive pattern recognition and language acquisition, this part of the system is equipped with a YOLO-based object detection model that has been trained on a dataset of objects designed for education such as coloured balls, boxes, and balloons. The system then creates adaptive pattern sequences depending on the detected objects, and these sequences are demonstrated by the robotic arm to assist the learner in identifying the visual patterns. Furthermore, these visual patterns are then transformed into verbal sequences using Google Voice Access while the response of the learner is collected, converted to text, and analysed using logistic regression classification.

In order to facilitate natural and assisted interaction, an intention recognition system based on gaze tracking is integrated into the system. The gaze images of the participants are recorded over seven predetermined directions, and a deep learning algorithm using ResNet-18 architecture is employed to train the model for classifying gaze directions. In the real-time application, the faces present in the environment are detected using a deep learning algorithm implemented through the Caffe framework, and head pose estimation techniques are employed to make the system more robust under natural conditions. Gaze direction estimates are smoothed out across subsequent frames, and the workspace grid is used to map the gaze and head poses to relevant objects.

The predictive sorting system is created to help students learn through observation and complete tasks. The workspace consists of various objects based on their forms (sphere, box, pyramid) and colours (red, green, blue). In addition, a detection model based on YOLOv8 detects the attributes of each object such as its colour, form, and location in the workspace. A comparison is performed between several machine learning models such as Random Forest, XGBoost, and Logistic Regression to develop a predictive algorithm that sorts objects efficiently, and the algorithm chosen was the Random Forest model since it was better at sorting objects in colour-based mode.

The robot arm incorporates all these elements into one system, which provides the user with a practical learning experience. The arm is controlled by an Arduino-based controller and works in a fixed space layout. It uses object recognition, eye tracking, and predictive sorting results to move around and retrieve or deliver objects to their correct places. It performs operations several times to allow observational learning and then lets the learners interact with it independently, allowing the learners to carry out sorting, replication of patterns, or object recognition operations.

4. Application Architecture and Technologies

This proposed system integrates technologies like robotics, computer vision, and deep learning to form a system that helps differently abled learners interactively learn. The use of a robotic arm allows for object handling and positioning, whereas an RGB camera can take snapshots of the targeted object in real-time. Preprocessing is done through the OpenCV and NumPy tools before frame processing and material classification. The proposed system makes use of DINOv2 self-supervised vision transformer implemented using PyTorch, which enables the system to extract visual features accurately from the captured frame without necessarily having to rely on vast labeled data for classification purposes of materials like plastics, metal, glass, and papers.

Object detection and exercises involving interaction are carried out using models built based on YOLO, with the training set consisting of custom-made images of classroom objects. Such machine learning models allow for real-time recognition and adaptive pattern generation needed in different learning tasks. In this regard, models of Machine Learning like Random Forest, XGBoost, and Logistic Regression suggest appropriate recommendations, whereas SHAP explainable artificial intelligence helps to make decisions understandable and clear. As far as verbal tasks go, Google Voice Access records students' responses that are analyzed using a logistic regression model. Thus, a combination of visual, verbal, and physical interactions is provided.

Gaze-based Human–Robot Interaction is achieved through a pre-trained deep-learning model Caffe used for face detection and ResNet-18 CNN model used for gaze estimation. The system guarantees efficient integration of all mentioned models, providing real-time processing, numerical computations, and system control. Commands are issued by means of the Arduino-controlled robot, which is used in combination with an IP webcam, allowing for seamless coordination of hardware and software modules.

With the use of modular design, it is easy to incorporate more applications in education and industry sectors. Through integration of robotics, computer vision, and deep learning, the system has shown that it can adapt through feedback, learning interactions, and explainability of its decision-making process.

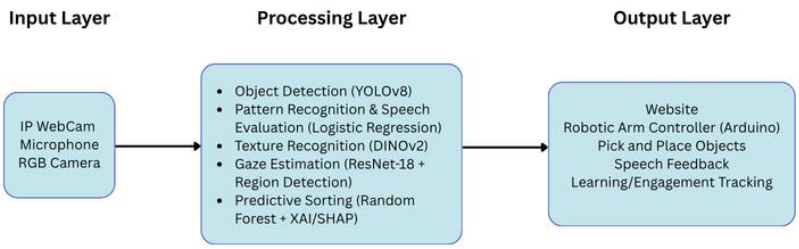

Fig. 1. System Architecture Diagram

The Fig. 1 depicts the three-tier structure of the suggested model, namely input, processing, and output. Input involves capturing real-time data through an IP webcam, RGB camera, and microphone. Within the processing phase, various AI models collaborate in the form of YOLOv8 for object recognition, ResNet-18 for eye gaze estimation, DINOv2 for feature recognition, Logistic Regression for pattern analysis, and Random Forest coupled with SHAP for decision-making and interpretation. Additionally, region detection plays a role in determining where users focus their attention and assigning these regions to corresponding zones. Output comprises executing certain activities based on the data processed, such as controlling a robotic arm for picking and placing operations, providing a web application for surveillance purposes, delivering speech feedback, and learning from users.

5. Results and Discussion

The texture recognition module was implemented with the aid of a self-supervised vision transformer DINOv2 fused with a customized PyTorch classifier model, which was trained using a dataset that included Glass, Metal, Plastic, Rubber, and Others. In order to assess the performance of the proposed method, other model architectures like Sequential CNN (Accuracy = 0.4419), EfficientNet (Accuracy = 0.8772), and MobileNet (Accuracy = 0.8872) were considered for comparison. It is evident from Table 4 that DINOv2 outperformed all other baseline methods by a wide margin, with a test accuracy of 0.98, and precision, recall, and F1-scores ranging from 0.95 to 0.99 in all categories.
More insights into the effectiveness of the model in discriminating between different materials based on their features can be seen from the t-SNE plots where it is clearly observable that all the embeddings of features belonging to a certain material class have been separated from other material classes despite differences in light conditions, orientation, and texture. The inference on unknown data yielded high confidence scores greater than 0.95, showing that this is indeed a reliable model for real-world applications. Using OpenCV, NumPy, DINOv2, and PyTorch for classification is highly efficient and interpretable.

Pattern learning through vision is performed through the use of YOLO object detection to ensure real-time object location and mapping of objects into predefined spatial areas. By detecting various objects within the environment, patterns are generated through a frequency-based matching method that will work for the detected objects, thus moving away from predefined fixed patterns toward contextually aware patterns. The generated pattern is partially shown to the user, while the last element is purposely concealed to generate a pattern completion problem. The robotic system performs the task in steps to help users learn in a guided manner.

Interaction within the robotic system is captured in a state-based approach that ensures actions of the robot and reactions of the user are recorded to create a learning loop in real time. In order to classify the reaction of users, Logistic Regression classifier was used, and its accuracy is 0.96, indicating reliable classification of interactions.
The performance of the gaze classification module was assessed on the basis of two models namely ResNet-18 architecture and MobileNet based architecture. Using ResNet-18 architecture, a test accuracy score of 86.50% with macro F1-score value equal to 0.8643 and a weighted F1-score of 0.8657 was obtained, thereby proving its consistent performance for all gaze categories. Excellent classification accuracy scores were obtained for well-defined gaze directions such as upper right (F1=0.9565) and center (F1=0.9500), which show the capacity of the proposed system to detect the gaze pattern effectively. For the ambiguous category of gaze, a relatively low F1 score (F1=0.7568) was obtained, which indicates that this type of gaze cannot be effectively classified.

On the other hand, MobileNet showed much lower performance values in terms of accuracy (33%) and macro F1-score (0.34). Such inferior performance can be attributed to the compact nature of the network architecture. The results suggest that ResNet-18 architecture offers much better feature extraction and classification ability compared to the other network architecture. Therefore, it can be used effectively for human-robot interaction applications based on intention-driven approach through gaze mapping to the predefined spatial location.

The object detection component, designed based on YOLOv8, is crucial in providing effective environmental perception. It has obtained scores of 0.949 for precision, 0.956 for recall, and 0.954 for mAP@50. This shows the model has performed well in detecting different objects including balls, boxes, and pyramids. Even though the model is relatively moderate when it comes to the mAP@50-95 criterion, with a score of 0.543, its performance is adequate for use in practical real-world robotic systems which do not need very precise location information. Furthermore, the model's quick inference time (2.8 ms per frame) and small size (6.3 MB) make it feasible for deployment.

After successful object detection, several models were compared in terms of sorting recommendation. The accuracy of the XGBoost model was 0.88, whereas Logistic Regression had the same score. In contrast to them, the Random Forest model performed better with about 0.94 accuracy that did not differ much regardless of the number of output classes (color - red, green, blue; shape – ball, box, pyramid). The use of a Multioutput Classifier was important in order to implement multi-target classification in the model. It allows for implementing the mode-based approach to sorting that considers frequency as a key criterion and improves overall efficiency.

The proposed design involves the fusion of YOLOv8 based object detection approach with the random forest recommendation algorithm in real time via the use of an IP camera and a robotically controlled setup using an Arduino device. Objects detected in real time are aligned in a grid pattern and undergo a process flow which sorts them and interacts with them according to specific criteria. The implementation of explainable AI via SHAP enables one to interpret the decision process used by the algorithm in an attempt to help educators understand the model and allow students to sort, compare and identify objects.

Table 1.Highlights of Discussion and Results

Component	Models Evaluated	Key Metrics	Selected Model
Texture Recognition	**DINOv2 (0.98),** Sequential CNN (0.4419), EfficientNet (0.8772), MobileNet (0.8872)	Accuracy: 0.98, Precision: 0.98, Recall: 0.98, F1-score: 0.98	DINOv2
Gaze Tracking	**ResNet-18 (0.865),** MobileNet (0.33)	Accuracy: 0.865, Loss: 0.7601, Macro F1: 0.8643, Weighted F1: 0.8657	ResNet-18
XAI based Adaptive Recommendation Sorting	(Object Detection) **YOLOv8**	Precision: 0.949, Recall: 0.956, mAP@50: 0.954, mAP@50–95: 0.543	YOLOv8
	(For Recommendations) **Random Forest (0.94),** XGBoost (0.88), Logistic Regression (0.88)	Accuracy: 0.94, Precision: 0.93, Recall: 0.74, F1-score: 0.82	Random Forest
Pattern Recognition	**Logistic Regression (0.9123),** Naive Bayes (0.88), decision tree (0.64)	Accuracy: 0.9123 Precision: 0.8750 Recall: 0.8200 F1-Score: 0.8467	Logistic Regression

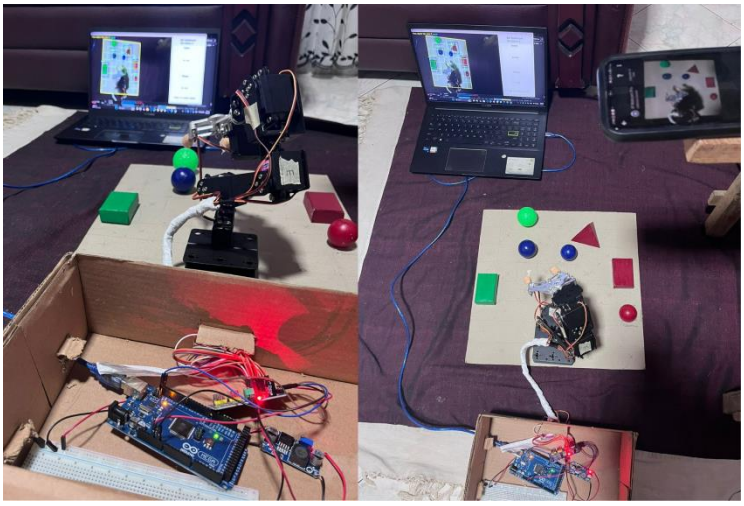

Fig. 2. System Setup of the Proposed Human–Robot Collaborative Learning Platform

The Fig.2. above depicts the system structure of the suggested human-robot collaboration learning framework. The robot arm works within a workspace that includes colored objects with varied shapes. The process is monitored through the use of an RGB camera mounted on the robot arm and connected to a smartphone. The laptop computer is responsible for processing the collected data for object recognition, gaze estimation, and region mapping while the robot arm movements are controlled by an Arduino-controlled board.

6. Conclusion and Future work

This work proposes a complete system of human-robot cooperative learning involving material recognition, visual pattern learning with speech interaction, gaze intention recognition, and predictive sorting using explainable AI. In material recognition using DINOv2 architecture, the module was able to achieve a high level of accuracy in classifying various materials, such as glass, metal, plastic, rubber, and other material types. Its accuracy rate was higher than alternative approaches used in other models, such as Sequential CNN, EfficientNet, and MobileNet. Visual pattern learning with speech interaction module was successful in combining object detection based on YOLO algorithm and response evaluation based on logistic regression for improving learner's cognitive abilities, ability to recognize patterns, and verbal skills, especially among learners with special needs. Using head pose estimation and temporal smoothing of ResNet-18 model, gaze-based interaction allowed for developing user-friendly interaction without requiring any special equipment. Moreover, the developed module was capable of providing consistent results in practical settings. Finally, predictive sorting using YOLOv8 detector, decision-making based on the Random Forest model, and SHAP explainability ensured accurate, efficient, and clear decision-making process.

For future work, there are several studies to enhance performance and scalability, as well as increase practical applicability of the proposed system. First, expanding the database used in training and validation in all system components material recognition, object identification, user gaze pattern analysis, and learning scenario generation would benefit from increasing the variety of environment conditions. Second, multimodal sensing techniques like depth cameras and tactile sensors might be implemented to make the system more robust and efficient. Attention-based architectures as well as continual learning methods should also be explored to deal with cases when the information is not sufficient and allow adaptation to new types of objects and users. In addition, personalization could be added to adjust the learning experience according to the specificities of each user. Moreover, implementation of optimizations in order to optimize for edge computing should also be considered as well as the extension of the application domain by adding dynamic environments and multiple robotic arms for collaboration. Lastly, conducting large-scale experiments with real users as well as establishing partnerships with institutions specializing in education and rehabilitation could play a vital role in successful commercialization of the proposed technology.

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